MACHINE LEARNING ENGINEER ASSESSMENT

Freight Rate Prediction

A leakage-aware forecasting pipeline for 12,000 future loads

Candidate	GitHub
eraj22	**github.com/eraj22**
Development data	Final scoring data
48,000 loads | Jan-Oct 2025	12,000 loads | Nov-Dec 2025

This report documents the data audit, chronological validation design, model choice, final holdout performance, and the required fixed December prediction chart.

Reproducible code | **Exact ID alignment** | **Scorer validated**

Executive summary

The final solution is a three-seed CatBoost ensemble trained on the logarithm of posted rate. It was selected because the dataset combines continuous signals, high-cardinality route categories, missing values, unseen lanes, nonlinear distance effects, and rare extreme rates.

$123.25	5.68%	0.820	51.4%
OCTOBER MAE	OCTOBER MAPE	OCTOBER R-SQUARED	MAE IMPROVEMENT

Decision summary

- Use a chronological split because the scoring rows occur after the labeled period.
- Use CatBoost to preserve categorical route information while remaining robust to unseen combinations.
- Predict log1p(posted_rate) to limit the leverage of rare rates above $10,000 without deleting them.
- Average three deterministic seeds to reduce model variance.
- Train the final ensemble on all 48,000 labeled rows after the October holdout was evaluated.

Validation design

Stage	Training window	Evaluation window	Purpose
Development	2025-01-01 to 2025-08-31	September 2025	Model and feature selection
Final holdout	2025-01-01 to 2025-09-30	October 2025	One-month forward simulation
Final fit	2025-01-01 to 2025-10-31	November-December 2025	Submission predictions

Why not random cross-validation? Random folds would mix future observations into past training windows. The chronological design better represents the deployment decision and provides a more credible estimate of forward performance.

Data audit

The labeled data contains 48,000 loads from January through October 2025. The submission data contains 12,000 loads from November through December. IDs are unique and the rows are not duplicated.

Check	Labeled	Final	Treatment
Duplicate rows	0	0	No action
Duplicate load_id	0	0	Hard validation failure
Missing weight	300	165	Missing flag and native null handling
Negative weight	292	145	Absolute value plus anomaly flag
Missing market index	374	249	Missing flag and native null handling
Unseen cities	-	8	Use coordinates and other numeric features
Unseen lanes	-	736	Do not rely on lane memorization

Target behavior

Posted rate is right-skewed. The median is $2,030.76, the mean is $2,373.98, the 99th percentile is $5,972.83, and the maximum is $25,533. Extreme values were retained because the brief does not identify them as bad records. A log target makes the model robust while preserving every row.

Generalization risk

Eight cities do not occur in the labeled data, and 736 final lanes are unseen. The feature set therefore combines lane identity with distance, coordinates, equipment, weight, market and quote signals, and calendar variables. An unseen lane can still be priced from its physical and market characteristics.

Leakage controls

Feature construction never uses posted_rate. All transformations are row-local or based only on training history. The fixed December scenario does not contain market index, quote signal, or coordinates, so coordinates are mapped from labeled city history and the two market signals use the most recent 30-day training medians. Those assumptions remain fixed while only the date changes.

Source: provided train_test.csv, validation.csv, and assessment brief. Audit counts are generated by the submitted pipeline.

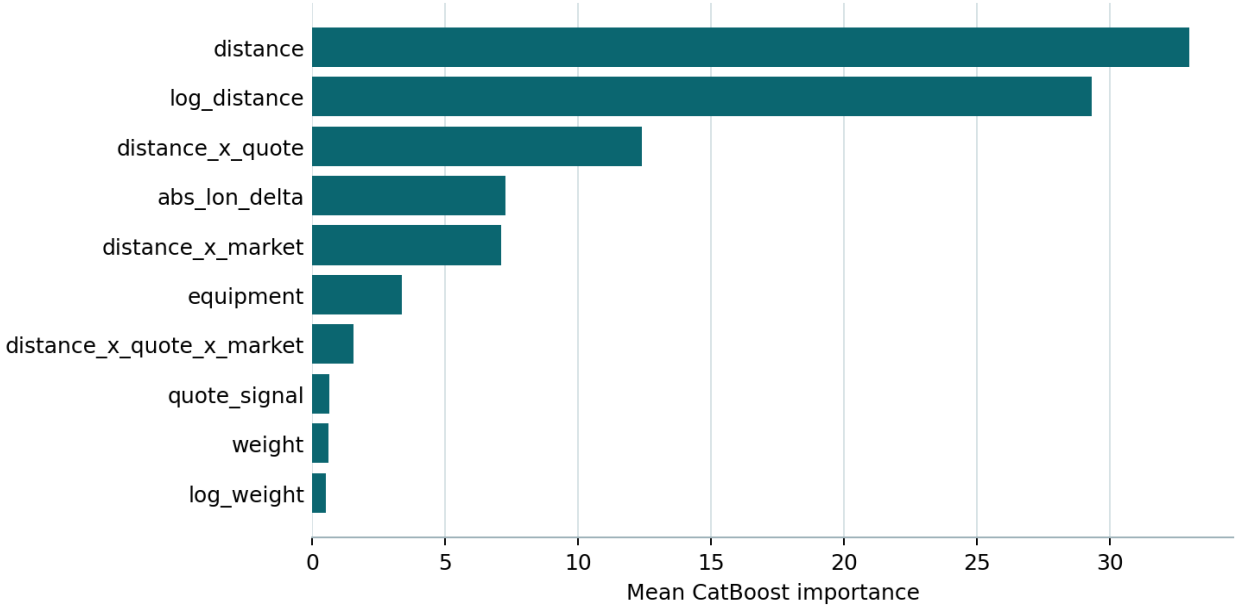
Model and feature engineering

CatBoost is a strong fit for this problem because it learns nonlinear interactions, processes categorical features directly, and handles missing numeric values. The final ensemble uses depth 8, 400 boosting iterations, learning rate 0.07, L2 regularization 8, and seeds 42, 137, and 2026.

Feature groups

Group	Examples	Reason
Route	pickup, delivery, lane, coordinates	Capture geographic and directional effects
Load	distance, equipment, cleaned weight	Represent the physical movement
Market	market index, quote signal	Represent observed price conditions
Interactions	distance x quote, distance x market	Convert per-mile signals into load-scale effects
Calendar	month, weekday, day of year, cyclic terms	Capture seasonal and weekly patterns
Quality flags	weight missing, negative weight, market missing	Preserve information about imperfect records

Top feature importance



Importance is averaged across the final three models. Distance and its nonlinear or market interactions dominate, while equipment and geographic displacement provide additional structure. Importance is descriptive, not causal.

Final holdout results

The final holdout contains 4,853 October loads that were not used to fit the evaluated models. The comparison baseline multiplies each distance by the median historical rate per mile from the training window.

Metric	Final ensemble	Rate-per-mile baseline	Change
MAE	$123.25	$253.85	-51.4%
RMSE	$647.62	$695.05	-6.8%
MAPE	5.68%	11.81%	-6.14 pp
Median absolute error	$39.73	$132.82	-70.1%
R-squared	0.820	0.793	+0.027

Error by equipment

Equipment	Rows	MAE	Median AE
Dry Van	2,745	$120.83	$40.02
Flatbed	918	$117.48	$36.85
Reefer	1,190	$133.29	$41.51

Interpretation

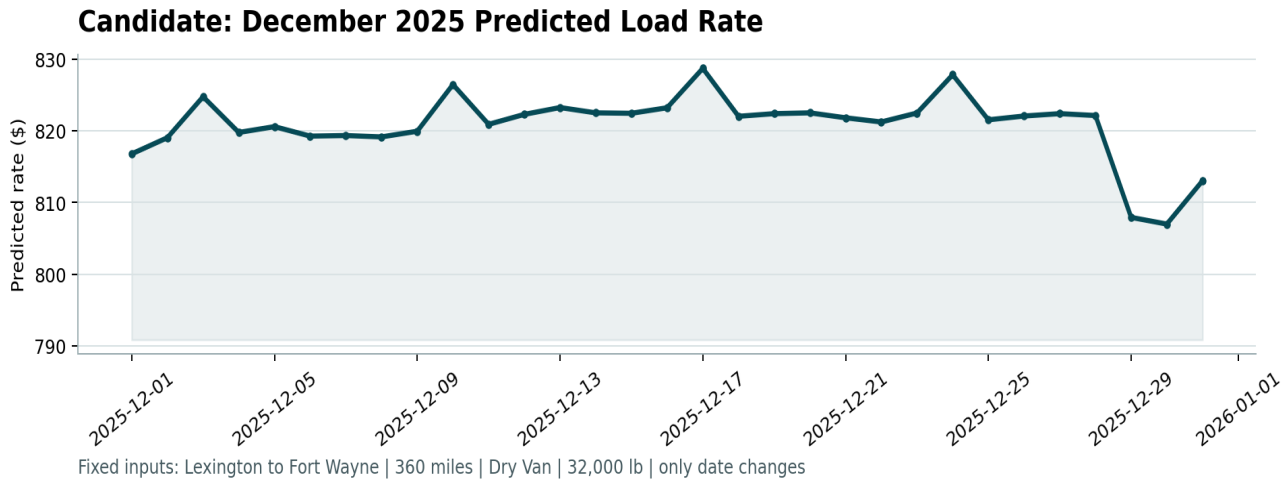
Typical-load accuracy is strong: the median absolute error is $39.73 and 90% of absolute errors are below $166.64. RMSE remains much larger than MAE because a small number of extreme rates create large squared errors. This gap supports the log-target design and continued human review for unusually expensive loads.

Reproducibility

The repository provides fixed dependency ranges, deterministic seeds, schema checks, output validation, unit tests, and GitHub Actions CI. The provided scorer successfully validates the 12,000 final IDs and all 31 fixed December rows.

Fixed December prediction

The chart below is generated by the employer-provided score.py after validating the completed December input file. Pickup, delivery, distance, equipment, and weight remain fixed. Only the date changes from 2025-12-01 through 2025-12-31.



Operational assumptions

- Lexington pickup and Fort Wayne delivery coordinates use median city coordinates from the labeled data.
- Missing December market index and quote signal use the latest 30-day medians available in the labeled data.
- The resulting date-only variation reflects learned calendar effects and does not assume knowledge of future market signals.

Production considerations

Monitor MAE, MAPE, and error by equipment and lane after actual rates arrive. Retrain when the market regime or feature distributions drift. Keep a manual review threshold for predictions outside the observed rate and distance ranges. The model supports pricing analysis; it should not replace commercial judgment for exceptional loads.

Repository run sequence

Step	Command
1	python -m pip install -r requirements.txt
2	python evaluate.py
3	python train_predict.py
4	python score.py --predictions validation_predictions.csv --december-predictions data/december_chart_inputs.csv
5	python -m unittest discover -s tests -v

Final artifacts: validation_predictions.csv, data/december_chart_inputs.csv, scorer_results/candidate_december.png, and this report.